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AEROSOL PROVISION SYSTEM AND METHOD

Abstract

An aerosol provision system comprises a haptic component and control circuitry. The control circuitry is configured to adjust a setting of the haptic component in response to determining that an inhalation on the aerosol provision system is about to occur.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to an aerosol provision system and method.

BACKGROUND

[0002] Electronic aerosol provision systems such as electronic cigarettes (e-cigarettes) generally contain an aerosol-generating material, such as a reservoir of a source liquid containing a formulation, typically including nicotine, or a solid material such as a tobacco-based product, from which an aerosol is generated for inhalation by a user, for example through heat vaporisation. Thus, an aerosol provision system will typically comprise an aerosol generator, e.g. a heating element, arranged to aerosolise a portion of aerosol-generating material to generate an aerosol in an aerosol generation region of an air channel through the aerosol provision system. As a user inhales on the device and electrical power is supplied to the aerosol generator, air is drawn into the device through one or more inlet holes and along the air channel to the aerosol generation region, where the air mixes with the vaporised aerosol generator and forms a condensation aerosol. The air drawn through the aerosol generation region continues along the air channel to a mouthpiece, carrying some of the aerosol with it, and out through the mouthpiece for inhalation by the user.

[0003] It is common for aerosol provision systems to comprise a modular assembly, often having two main functional parts, namely an aerosol provision device and an article. Typically, the article will comprise the article aerosol-generating material and the aerosol generator (heating element), while the aerosol provision device part will comprise longer-life items, such as a rechargeable battery, device control circuitry and user interface features. The aerosol provision device may also be referred to as a reusable part or battery section and the article may also be referred to as a consumable, disposable/replaceable part, cartridge or cartomiser.

[0004] The aerosol provision device and article are mechanically coupled together at an interface for use, for example using a screw thread, bayonet, latched or friction fit fixing. When the aerosol-generating material in an article has been exhausted, or the user wishes to switch to a different article having a different aerosol-generating material, the article may be removed from the aerosol provision device and a replacement article may be attached to the device in its place.

[0005] Haptic components or elements can form a part of the aerosol provision system and be used to provide a sensory signal or sensation to the user. Determining how and when to provide such a signal such that the user is able to detect and interpret/understand the signal can be challenging.

[0006] Various approaches are described herein which seek to help address or mitigate some of the issues discussed above.

SUMMARY

[0007] The disclosure is defined in the appended claims.

[0008] In accordance with some embodiments described herein, there is provided an aerosol provision system comprising a haptic component and control circuitry. The control circuitry is configured to adjust a setting of the haptic component in response to determining that an inhalation on the aerosol provision system is about to occur.

[0009] The aerosol provision system can further comprise one or more sensors, and wherein the control circuitry is configured determine the inhalation is about to occur based on signals received from one or more sensors. One of the sensors can be configured to detect if the aerosol generating system is connected to an external power source. One of the sensors can be configured to measure ambient noise around the aerosol provision system. One of the sensors can be configured to measure ambient light around the aerosol provision system. One of the sensors can be configured to measure an orientation and/or movement of the aerosol provision system. One of the sensors can be configured to detect if the user is touching or within proximity of the aerosol provision system.

[0010] The aerosol provision system can further comprise a mouthpiece. One of the sensors can be configured to detect if the user is touching or within proximity of the mouthpiece. One of the sensors can be configured to detect a pressure change at the mouthpiece.

[0011] Adjusting the setting of the haptic component can correspond to enabling the haptic component.
[0012] The setting can be an amplitude and/or frequency of a vibration generated by the haptic component. The setting can be a magnitude of a force generated by the haptic component.
[0013] The aerosol provision system can further comprise a display component, and the control circuitry is further configured to adjust a setting of the display component based on the determined orientation. The setting can be a brightness of the display component. Adjusting the setting of the display component can correspond to disabling the display component.
[0014] The aerosol provision system can further comprise a speaker component, and the control circuitry is further configured to adjust a setting of the speaker component based on the determined orientation. The setting can be a volume of the speaker component. Adjusting the setting of the speaker component can correspond to disabling the speaker component.
[0015] The aerosol provision system can further comprise a communications interface, and wherein control circuitry is further configured to adjust a setting of the haptic component based on a signal received via the communications interface from an external device.
[0016] The aerosol provision system can further comprise an input device, and wherein control circuitry is further configured to adjust a setting of the haptic component based on an input received via the input device.
[0017] In accordance with some embodiments described herein, there is provided an aerosol provision device for an aerosol provision system, the aerosol provision system comprising a haptic component, wherein the aerosol provision device comprises control circuitry configured to adjust a setting of the haptic component in response to determining that an inhalation on the aerosol provision system is about to occur.
[0018] In accordance with some embodiments described herein, there is provided a method for operating an aerosol provision system comprising adjusting a setting of a haptic component of the aerosol provision system in response to determining that an inhalation on the aerosol provision system is about to occur.
[0019] There is also provided a computer readable storage medium comprising instructions which, when executed by a processor, performs the above method.
[0020] These aspects and other aspects will be apparent from the following detailed description. In this regard, particular sections of the description are not to be read in isolation from other sections.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0021] Embodiments of the invention will now be described, by way of example only, with reference to accompanying drawings, in which:
[0022] FIG. **1** is a schematic diagram of an aerosol provision system;
[0023] FIG. **2** is a flow diagram of a method for operating an aerosol provision system.
DETAILED DESCRIPTION
[0024] Aspects and features of certain examples and embodiments are discussed/described herein. Some aspects and features of certain examples and embodiments may be implemented conventionally and these are not discussed/described in detail in the interests of brevity. It will thus be appreciated that aspects and features of articles and systems discussed herein which are not described in detail may be implemented in accordance with any conventional techniques for implementing such aspects and features.
[0025] The present disclosure relates to aerosol provision systems, which may also be referred to as vapour provision systems, such as e-cigarettes. Throughout the following description the term “e-cigarette” or “electronic cigarette” may sometimes be used, but it will be appreciated this term may

be used interchangeably with aerosol provision system and electronic aerosol provision system.

[0026] As noted above, aerosol provision systems (e-cigarettes) often comprise a modular assembly including both a reusable part (aerosol provision device) and a replaceable (disposable) or refillable cartridge part, referred to as an article. Systems conforming to this type of two-part modular configuration may generally be referred to as two-part systems or devices. It is also common for electronic cigarettes to have a generally elongate shape. For the sake of providing a concrete example, certain embodiments of the disclosure described herein comprise this kind of generally elongate two-part system employing refillable cartridges. However, it will be appreciated the underlying principles described herein may equally be adopted for other electronic cigarette configurations, for example modular systems comprising more than two parts, as devices conforming to other overall shapes, for up example based on so-called box-mod high performance devices that typically have a more boxy shape, or even systems comprising one part where the aerosol provision device and article are integrally formed with one another.

[0027] FIG. **1** is a highly schematic diagram (not to scale) of an example aerosol provision system **10**, such as an e-cigarette, to which embodiments are applicable. The aerosol provision system **10** has a generally cylindrical shape, extending along a longitudinal or y axis as indicated by the axes (although aspects of the invention are applicable to e-cigarettes configured in other shapes and arrangements), and comprises two main components, namely an aerosol provision device **20** and an article **30**.

[0028] The article **30** comprises or consists of aerosol-generating material **32**, part or all of which is intended to be consumed during use by a user. An article **30** may comprise one or more other components, such as an aerosol-generating material storage area **39**, an aerosol-generating material transfer component **37**, an aerosol generation area, a housing, a wrapper, a mouthpiece **35**, a filter and/or an aerosol-modifying agent.

[0029] An article **30** may also comprise an aerosol generator **36**, such as a heating element, that emits heat to cause the aerosol-generating material **32** to generate aerosol in use. The aerosol generator **36** may, for example, comprise combustible material, a material heatable by electrical conduction, or a susceptor. It should be noted that it is possible for the aerosol generator **36** to be part of the aerosol provision device **20** and the article **30** then may comprise the aerosol-generating material storage area **39** for the aerosol-generating material **32** such that, when the article **30** is coupled with the aerosol provision device **20**, the aerosol-generating material **32** can be transferred to the aerosol generator **36** in the aerosol provision device **20**. It should be appreciated that the aerosol generator **36** may encompass an aerosol generator other than a heater. More generally, an aerosol generator is an apparatus configured to cause aerosol to be generated from the aerosol-generating material. In some other embodiments, the aerosol generator is configured to cause an aerosol to be generated from the aerosol-generating material without heating. For example, the aerosol generator may be configured to subject the aerosol-generating material to one or more of vibration, increased pressure, or electrostatic energy.

[0030] Aerosol-generating material is a material that is capable of generating aerosol, for example when heated, irradiated or energized in any other way. The aerosol-generating material **32** may, for example, be in the form of a solid, liquid or gel which may or may not contain an active substance and/or flavourants. In some embodiments, the aerosol-generating material **32** may comprise an “amorphous solid”, which may alternatively be referred to as a “monolithic solid” (i.e. non-fibrous). In some embodiments, the amorphous solid may be a dried gel. The amorphous solid is a solid material that may retain some fluid, such as liquid, within it. In some embodiments, the aerosol-generating material **32** may for example comprise from about 50 wt %, 60 wt% or 70 wt % of amorphous solid, to about 90 wt %, 95 wt% or 100 wt % of amorphous solid.

[0031] The aerosol-generating material comprises one or more ingredients, such as one or more active substances and/or flavourants, one or more aerosol-former materials, and optionally one or more other functional materials such as pH regulators, colouring agents, preservatives, binders,

fillers, stabilizers, and/or antioxidants.
[0032] The active substance as used herein may be a physiologically active material, which is a material intended to achieve or enhance a physiological response. The active substance may for example be selected from nutraceuticals, nootropics, and psychoactives. The active substance may be naturally occurring or synthetically obtained. The active substance may comprise for example nicotine, caffeine, taurine, theine, vitamins such as B6 or B12 or C, melatonin, cannabinoids, or constituents, derivatives, or combinations thereof. The active substance may comprise one or more constituents, derivatives or extracts of tobacco, cannabis or another botanical.
[0033] In some embodiments, the active substance comprises nicotine. In some embodiments, the active substance comprises caffeine, melatonin or vitamin B12.
[0034] The aerosol provision device **20** includes a power source **14**, such as a battery, configured to supply electrical power to the aerosol generator **36**. The power source **14** in this example is rechargeable and may be of a conventional type, for example of the kind normally used in electronic cigarettes and other applications requiring provision of relatively high currents over relatively short periods. The power source **14** may be recharged through the charging port (not illustrated), which may, for example, comprise a USB connector.
[0035] The aerosol provision device **20** includes device control circuitry **28** configured to control the operation of the aerosol provision system **10** and provide conventional operating functions in line with the established techniques for controlling aerosol provision systems such as electronic cigarettes. The device control circuitry (processor circuitry) **28** may be considered to logically comprise various sub-units/circuitry elements associated with different aspects of the electronic cigarette's operation. For example, depending on the functionality provided in different implementations, the (device) control circuitry **28** may comprise power source control circuitry for controlling the supply of electrical power from the power source **14** to the aerosol generator **36**, user programming circuitry for establishing configuration settings (e.g. user-defined power settings) in response to user input, as well as other functional units/circuitry associated functionality in accordance with the principles described herein and conventional operating aspects of electronic cigarettes. It will be appreciated the functionality of the (device) control circuitry **28** can be provided in various different ways, for example using one or more suitably programmed programmable computer(s) and/or one or more suitably configured application-specific integrated circuit(s)/circuitry/chip(s)/chipset(s) configured to provide the desired functionality.
[0036] The aerosol provision device **20** has an interface configured to receive the article **30**, thereby facilitating the coupling between the aerosol provision device **20** and the article **30**. The interface is located on a surface of the aerosol provision device **20**.
[0037] The housing of the article **30** has a surface configured to be received by the interface on the aerosol provision device **20** in order to facilitate coupling between the article **30** and the aerosol provision device **20**. The surface of the article may be configured to be a size and/or shape that mirrors the size and/or shape of the interface in order to facilitate coupling between the aerosol provision device **20** and the article **30**. For example, the interface may comprise a cavity, chamber or other space on the surface of the aerosol provision device **20**. The surface of the article **30** can then be configured to be a size and shape that mirrors the size and shape of the cavity in order for the surface of the article **30** to be inserted into the cavity.
[0038] Although not illustrated, the interface of the aerosol provision device **20** and the surface of the article **30** may have complementary features to reversibly attach and mate the article **30** to the aerosol provision device **20**, such as a screw thread, bayonet fitting, latched or friction fit fixing or other fastening means.
[0039] The interface also comprises one or more connectors, such as contact electrodes, connected via electrical wiring to the control circuitry **28** and the power source **14**. The article **30** also comprises one or more connectors, such as contact electrodes, connected via electrical wiring to the aerosol generator **36**. In use, the article **30** is received by the interface of the aerosol provision

device **20**, thereby coupling the aerosol provision device **20** and the article **30**. This results in the connectors on the article **30** mating with the connectors on the aerosol provision device **20**, thereby allowing electrical power and electrical current to be supplied from the power source **14** of the aerosol provision device **20** to the aerosol generator **36** of the article **30**.

[0040] The housing of the article **30** has a surface configured to engage with an interface on the aerosol provision device **20** in order to facilitate coupling between the article **30** and the aerosol provision device **20**. In other words, the aerosol provision device **20** is configured to receive the article **30**, via the interface, and the surface of the article is proximate to the interface on the aerosol provision device **20** when the article **20** is received by the interface.

[0041] The aerosol provision system **10** includes one or more air inlets **21**, located on one or more of the aerosol provision device **20** and the article **30**. In use, as a user inhales on the mouthpiece **35**, air is drawn into the aerosol provision system **10** through the air inlets **21** and along an air channel **23** to the aerosol generator **36**, where the air mixes with the vaporised aerosol-generating material **32** and forms a condensation aerosol. The air drawn through the aerosol generator **36** continues along the air channel **23** to a mouthpiece **35**, carrying some of the aerosol with it, and out through the mouthpiece **35** for inhalation by the user.

[0042] By way of a concrete example, the article **30** comprises a housing (formed, e.g., from a plastics material), an aerosol-generating material storage area **39** formed within the housing for containing the aerosol-generating material **32** (which in this example may be a liquid which may or may not contain nicotine), an aerosol-generating material transfer component **37** (which in this example is a wick formed of e.g., glass or cotton fibres, or a ceramic material configured to transport the liquid from the reservoir using capillary action), an aerosol-generating area containing the aerosol generator **36**, and a mouthpiece **35**. Although not shown, a filter and/or aerosol modifying agent (such as a flavour imparting material) may be located in, or in proximity to, the mouthpiece **35**. The aerosol generator **36** of this example comprises a heater element formed from an electrically resistive material (such as NiCr8020) spirally wrapped around the aerosol-generating material transfer component **37**, and located in an air channel **23**. The area around the heating element and wick combination is the aerosol-generating area of the article **30**.

[0043] As illustrated in FIG. **1**, the aerosol provision system **10** also comprises a haptic component **22**. The haptic component **22** is configured to generate one or more vibrations and/or forces in order to provide a haptic sensation to the user of the aerosol provision system **10**. The haptic component **22** can comprise any suitable component configured to provide a haptic sensation (feedback) to a user. For example, the haptic component **22** may comprise an eccentric rotating mass (ERM) or linear resonant actuator (LRA) haptic component. The haptic component **22** may comprise a piezoelectric actuator configured to produce one more vibrations, and/or the haptic component **22** can comprise a motor configured to produce one or more forces. For example, a pneumatic motor, valve and/or fan can be used to generate a puff, pulse or other movement of air that may be sensed by the user of the aerosol provision system **10**. Additionally, or alternatively, the haptic component **22** may generate haptic sensations that do not directly result from a motion (e.g., vibration) of a mass or of air; e.g., the haptic component **22** may be configured to generate a haptic sensation via an electromagnetic field or via an electric pulse, for example.

[0044] The haptic component **22** can be located on or within the aerosol provision device **20** and/or the article **30**. For example, the haptic component **22** can be located proximate to or forming part of the housing of the aerosol provision device **20** such that the vibrations and/or forces generated by the haptic component **22** can be felt by the user of the aerosol provision system **10** when the user of the aerosol provision system **10** is holding the aerosol provision device **20**. The aerosol provision system **10** may comprise more than one haptic component **22**, for example one located proximate to or forming part of the housing of the aerosol provision device **20** and one located proximate to or forming part of the article **30**, such as on the mouthpiece **35** such that the vibrations/forces generated by the haptic component **22** are sensed on the lips of the user during an inhalation.

[0045] The haptic component **22** can be operatively coupled to the power source **14** of the aerosol provision device **20** in order to receive electrical power, and the haptic component **22** can be operatively coupled to the control circuitry **28** such that the control circuitry **28** can be configured to control the haptic component **22**.
[0046] The control circuitry **28** is configured to adjust a setting of the haptic component **22** in response to determining that an inhalation on the aerosol provision system is about to occur. As described above, in order to use the aerosol provision system **10**, the user inhales, sucks or otherwise draws on the aerosol provision system **10** in order to cause aerosol to be generated and delivered into the mouth of the user. The control circuitry **28** is configured to determine that such an inhalation on the aerosol provision system is about to occur. In other words, the control circuity **28** is configured to anticipate or otherwise predict when an inhalation is going to occur.
[0047] For example, the aerosol provision system **10** may comprise a button, switch or other form of input device which the user is required hold or actuate in order to perform an inhalation on the aerosol provision system **10**. The input device may be connected to the aerosol generator **36** such that the aerosol generator **36** is only enabled when the input device is held or actuated by the user. The control circuitry **28** can be configured to determine the inhalation is about to occur based on an input on the input device.
[0048] Equally, the control circuitry **28** can be configured determine the inhalation is about to occur based on signals received from the one or more sensors. The aerosol provision system **10** illustrated in FIG. **1** comprises one or more sensors **24**. The one or more sensors **24** can be located on or within the aerosol provision device **20** and/or the article **30**. The one or more sensors **24** are operatively coupled to the power source **14** of the aerosol provision device **20** in order to receive electrical power, and the one or more sensors **24** are operatively coupled to the control circuitry **28** such that the control circuitry **28** can be configured to control the one or more sensors **24**. The control circuity **28** is then configured to determine the inhalation is about to occur based on signals received from the one or more sensors **24**. In other words, the control circuit **28** is configured to read or otherwise receive readings from the one or more sensors **24**, and use the readings to determine whether an inhalation is about to occur. The control circuitry **28** can receive readings from the sensors **24** periodically, for example every second, minute or five minutes, or the control circuitry **28** can send a request to the sensors **24** for the one or more readings. Equally, the control circuitry **28** can be configured to determine whether an inhalation is about to occur in response to an event, for example one or more readings from the sensors **24** changing.
[0049] One of the sensors **24** can be configured to detect if the aerosol generating system **10** is connected to an external power source. For example, one of the sensors **24** can comprise a current sensor configured to measure an amount of current received from an external power supply. When the power source or battery **14** of the aerosol provision system **10** needs to be recharged, the aerosol provision system **10** can be connected to an external power supply in order to recharge the power source or battery **14**. The current sensor can be located between the power source **14** and charging port of the aerosol provision system **10** in order to measure current flow between the external power supply and the power source **14** (i.e. to measure or otherwise detect when the aerosol provision system **10** is connected to an external power supply). It will be appreciated that users typically do not inhale on an aerosol provision system **10** whilst it is being recharged (i.e. whilst aerosol provision system **10** is connected to the external power supply).
[0050] One of the sensors **24** can be configured to measure ambient noise around the aerosol provision system **10**. For example, one of the sensors **24** can comprise microphone configured to measure the amount of noise proximate to the aerosol provision system **10**. Equally, the microphone or other noise sensor could be located proximate to the air inlets **21** and/or air channel **23** in order to detect noise associated with the flow of air into and through the air channel **23**. The sensor is then able to detect when air begins to flow into and through the air channel **23**, thereby indicating that the user is about to start or begin an inhalation. Additionally or alternatively, the

microphone or other noise sensor may be configured to detect changes of the airflow into and through the air channel **23** resulting from movement of the aerosol provision system **10**. For example, this airflow may be from the aerosol provision system **10** being moved towards the user's mouth. Additionally or alternatively, the microphone or other noise sensor may be configured to detect a sound resulting from a user's physical interaction with the aerosol provision system, e.g., the sound of a button click, the sound of the user gripping the aerosol provision system **10**, etc.

[0051] One of the sensors **24** can be configured to measure ambient light around the aerosol provision system **10**. For example, one of the sensors **24** can comprise an optical sensor configured to measure the amount of light proximate to the aerosol provision system **10**. .sub.[MB(1]When the user is holding the aerosol provision system **10** and about to inhale on the aerosol provision system **10**, the user's hand, lips or other part of the user's body may partially or completely obscure the optical sensor, thereby reducing the amount of light proximate to the aerosol provision system and detected by the optical sensors. The optical sensor can be located on a portion of the aerosol provision system **10**, such as the body of the aerosol provision device **20** or the mouthpiece **35**, where a user typically holds or touches the aerosol provision system **10** such that the amount of light detected by the optical sensors is indicative of whether or not the user is holding or touching the aerosol provision system **10**, since the user needs to touch the aerosol provision system **10** in order to perform an inhalation.

[0052] One of the sensors **24** can be configured to detect if the user is touching the aerosol provision system. For example, one of the sensors **24** can comprise a capacitive sensor, pressure sensor or other form of touch sensor. Such a touch sensor can be located on the body of the aerosol provision system **10**, for example on the aerosol provision device **20**, in order to detect if the user is holding or otherwise touching the aerosol provision system **10** with their hand. Equally, as described above, the aerosol provision system **10** can comprise a mouthpiece **35**, and the one of the sensor **24** can be configured to detect if the user is touching the mouthpiece **35**. For example, one of the sensors **24** a touch sensor located on the mouthpiece **35** and configured to detect if the user is touching the mouthpiece (e.g. with their lips or other part of their mouth), since the user needs to touch the aerosol provision system **10** in order to perform an inhalation. In some implementations, the one of the sensors **24** may be configured to detect whether the user is in proximity of the aerosol provision system **10**. For example, some capacitive sensors may be operated to detect a hand (or conductive object) a few centimetres from the surface of the capacitive touch sensor.

[0053] One of the sensors **24** can be configured to detect a pressure change at the mouthpiece **35**. For example, one of the sensors **24** can comprise pressure sensor located in or proximate to the exit of the air channel **23** through the mouthpiece such that a pressure change at the mouthpiece can be detected. Equally, the pressure sensor can be located on or proximate to the mouthpiece and configured to detect a pressure change at the mouthpiece outside of the air channel **23**, for example a pressure change associate with the user putting the mouthpiece in their mouth. As will be appreciated, before beginning an inhalation, the user will place at least the mouthpiece **35** of the aerosol provision system **10** into their mouth and form a seal on the aerosol provision system **10** with their lips. This will result in a change in air pressure, which one of the sensors **24** can be configured to detect.

[0054] One of the sensors **24** can be configured to measure an orientation and/or movement of the aerosol provision system **10**. For example, one of the sensors **24** can comprise a gyroscope, such as the microelectromechanical systems (MEMS) gyroscope or gyrometer, a 6 or 9 axis accelerometer, a MEMS accelerometer and/or an Inertial Measurement Unit (IMU). The orientation of the aerosol provision system **10** measured by the accelerometer can indicate whether the aerosol provision system **10** is located on a horizontal flat surface (i.e. along the x-axis in FIG. **1**), or whether the aerosol provision system **10** is in a different orientation such as with the article **30** pointing substantially downwards (i.e. with the mouthpiece **35** being at the lowest point on the aerosol provision system **10**, in opposite orientation in the y-direction to that illustrated in FIG. **1**). The accelerometer can also be configured to measure the motion of the aerosol provision system **10**, and therefore detect whether the aerosol provision system **10** is stationary, moving at a constant velocity, or undergoing an acceleration or deceleration. In some implementations, determining that an inhalation on the aerosol provision system is about to occur involves determining when a pattern of motion is detected (for example, corresponding to the motion of raising the aerosol provision system **10** to the user's mouth).

[0055] In response to determining that an inhalation on the aerosol provision system is about to occur, the control circuitry **28** is configured to adjust the setting of the haptic component **22**. The setting can be one or more of a duration, magnitude, amplitude, frequency and/or pattern of the one or more vibrations or forces generated by the haptic component. For example, the setting can be an amplitude and/or frequency of a vibration generated by the haptic component. The amplitude and/or frequency of the vibration can correspond to a plurality of vibrations, which may form a pattern of vibrations (in other words, one or more adjustments to the amplitude and/or frequency of vibration over time. The pattern might correspond to the haptic component being on (or a vibration being produced) for 1 second, off/stopped for two second, on/produced again for 1 second, or any other suitable pattern involving multiple vibrations over time. Since the aerosol provision component **10** is held by the user during the inhalation, the adjustment to the setting of the haptic component made by the control circuitry **28** will be sensed by the user. This haptic sensation will then be felt during the inhalation, changing the overall sensation of the inhalation and using the aerosol provision system **10**, thereby improving the user experience. In some implementations, the haptic sensation during an inhalation may be set so as to be more easily perceived by a user (e.g., the haptic sensation may be relatively stronger). This may relatively increase the level of haptic sensation that the user feels during inhalation acting to modify or enhance the user's experience when inhaling aerosol. In other implementations, the haptic sensation during an inhalation may be set so as to be less easily perceived by a user (e.g., the haptic sensation may be relatively weaker). This may relatively decrease the level of haptic sensation that the user feels during the inhalation meaning any haptic sensation may not detract (or detracts less) from the user's experience. The relative strength of the haptic sensation during an inhalation may be set in view of a particular user experience to be provided.

[0056] Equally, the setting can be a duration and/or magnitude of a force generated by the haptic component **22**. Adjusting the setting of the haptic component **22** can also correspond to enabling the haptic component **22**. In other words, in response to determining that an inhalation on the aerosol provision system is about to occur, the control circuitry **28** can be configured to enable the haptic component **22**. For example, the control circuitry **28** can be configured to allow or otherwise enable the supply of electrical power from the power source **14** to the haptic component **22**. Equally, adjusting the setting of the haptic component **22** can also correspond to disabling the haptic component **22**, for example by preventing the supply of electrical power from the power source **14** to haptic component **22**.

[0057] The control circuitry **28** can also be configured determine that the inhalation on the aerosol provision system **10** has finished or otherwise been completed in a similar fashion to determining that the inhalation was about to occur, for example based on signals received from the one or more sensors **24** or based on an input on the input device. In response to determining that the inhalation on the aerosol provision system has finished, the control circuitry **28** can be configured to adjust the setting of the haptic component **22** again, for example by undoing or reverse the setting adjustment that was made in response to determining that the inhalation on the aerosol provision system is about to occur. For example, if the amplitude of the vibration generated by the haptic component **22** was increased in response to determining that the inhalation was about to occur, the amplitude of the vibration generated by the haptic component **22** can be decreased in response to determining that the inhalation has finished. Equally, where the haptic component **22** was enabled in response to determining that the inhalation was about to occur, the haptic component **22** can be disabled in response to determining that the inhalation has finished.

[0058] Although not illustrated in FIG. **1**, the aerosol provision system **10** may comprise one or more output components, such as a display component and/or a speaker component. The control circuitry **28** can be configured to adjust a setting of the display component and/or the speaker component in response to determining that an inhalation on the aerosol provision system is about to occur.

[0059] In the case of the display component, the setting of the display component can correspond to a brightness, contrast, colour balance or other visual property of the display component (i.e. the screen of the display component). In the case of the speaker component, the setting of the speaker component can correspond to a volume, balance or other auditory property of the speaker component. Equally, adjusting the setting of the display device and/or speaker component can correspond, respectively, to enabling or disabling the display component and/or speaker component. Adjusting the setting of the display component and/or the speaker component could occur substantially simultaneously with adjusting the setting of the haptic component **22**, or in response to adjusting the setting of the haptic component **22**. For example, the control circuitry **28** could be configured to adjust the brightness of the display component (e.g. lower the brightness) and/or adjust the volume of the speaker component (e.g. lower the volume) in response to enabling the haptic component **22**. Equally, the control circuitry **28** could be configured to disable the display component and/or the speaker component when the haptic component **22** is enabled, then enable the display component and/or the speaker component when the haptic component **22** is disabled. This allows a primary component for conveying signals to the user to be changed between the one or more output components and the haptic component **22**.

[0060] The aerosol provision system **10** can also comprise a communications interface, and the control circuitry **28** configured to communicate with one or more external devices, such as a computer, mobile phone or other electronic device, via the communications interface. The communications interface can be configured to communicate using a suitable wireless communications protocol such as Wi-Fi, Bluetooth, RFID, NFC. The control circuitry **28** can be configured to adjust a setting of the haptic component **22**, such as the settings described above, based on signal received via the communications interface from an external device. In other words, the external device can send a signal to the aerosol provision system **10** by the communications interface, and in response to receiving the signal, the control circuit **28** adjusts a setting of the haptic component **22**. This allows the setting of the haptic component **22** to be adjusted without directly interacting with the aerosol provision system **10**. The signal received from the external device via the communications interface may indicate the setting and/or magnitude or type of adjustment to be made to the setting of the haptic component **22** to be adjusted, or the control circuitry **28** may be configured to determine the setting of haptic component **22** to adjust based on the signal received from the external device.

[0061] The aerosol provision system **10** can also comprise an input device, such as a button, switch, or touchscreen display. In the latter case, the display component described above as an output component can be a touchscreen component that is also configured to receive touch inputs. The control circuitry can be configured to adjust a setting of the haptic component **22**, such as the settings described above, based on an input received via the input device. In other words, the user can provide an input on the input device, and in response to receiving the input, the control circuit **28** adjusts a setting of the haptic component **22**. This allows the setting of the haptic component **22** to be adjusted by the user providing an input on the aerosol provision system **10**. The input received from the input device (e.g. from the user) may indicate the setting and/or magnitude or type of adjustment to be made to the setting of the haptic component **22** to be adjusted, or the control circuitry **28** may be configured to determine the setting of haptic component **22** to adjust based on the input received from the input device.

[0062] FIG. **2** is a flow diagram of a method **200** for operating an aerosol provision system, such as aerosol provision system **10**. The method begins at step **210**, where it is determined that an inhalation on the aerosol provision system is about to occur. At step **220**, a setting of a haptic component of the aerosol provision system is adjusted. The method then ends.

[0063] The method **200** illustrated in FIG. **2** may be stored as instructions on a computer readable storage medium, such that when the instructions are executed by a processor, the method described above is performed. The computer readable storage medium may be non-transitory. In other words, the method **200** illustrated in FIG. **2** may be computer implemented. The method **200** may be performed by the aerosol provision device **20**, such as by the control circuitry **28**.

[0064] As described above, the present disclosure relates to (but it not limited to) an aerosol provision system comprising a haptic component and control circuitry. The control circuitry is configured to adjust a setting of the haptic component in response to determining that an inhalation on the aerosol provision system is about to occur.

[0065] Thus, there has been described an aerosol provision system and method.

[0066] The various embodiments described herein are presented only to assist in understanding and teaching the claimed features. These embodiments are provided as a representative sample of embodiments only, and are not exhaustive and/or exclusive. It is to be understood that advantages, embodiments, examples, functions, features, structures, and/or other aspects described herein are not to be considered limitations on the scope of the invention as defined by the claims or limitations on equivalents to the claims, and that other embodiments may be utilised and modifications may be made without departing from the scope of the claimed invention. Various embodiments of the invention may suitably comprise, consist of, or consist essentially of, appropriate combinations of the disclosed elements, components, features, parts, steps, means, etc., other than those specifically described herein. In addition, this disclosure may include other inventions not presently claimed, but which may be claimed in future.

Claims

1. An aerosol provision system comprising: a haptic component; and control circuitry configured to adjust a setting of the haptic component in response to determining that an inhalation on the aerosol provision system is about to occur.

2. The aerosol provision system of claim 1, further comprising one or more sensors, and wherein the control circuitry is configured determine the inhalation is about to occur based on signals received from the one or more sensors.

3. The aerosol provision system of claim 2, wherein one of the sensors is configured to detect if the aerosol generating system is connected to an external power source.

4. The aerosol provision system of claim 2, wherein one of the sensors is configured to measure ambient noise around the aerosol provision system.

5. The aerosol provision system of claim 2 **4**, wherein one of the sensors is configured to measure ambient light around the aerosol provision system.

6. The aerosol provision system of claim 2, wherein one of the sensors is configured to measure an orientation and/or movement of the aerosol provision system.

7. The aerosol provision system of claim 2, wherein one of the sensors is configured to detect if the user is touching or within proximity of the aerosol provision system.

8. The aerosol provision system of claim 2, further comprising a mouthpiece.

9. The aerosol provision system of claim 8, wherein one of the sensors is configured to detect if the user is touching or within proximity of the mouthpiece.

10. The aerosol provision system of claim 8, wherein one of the sensors is configured to detect a pressure change at the mouthpiece.

11. The aerosol provision system of claim 1, wherein adjusting the setting of the haptic component corresponds to enabling the haptic component.

12. The aerosol provision system of claim 1, wherein the setting is an amplitude and/or frequency of a vibration generated by the haptic component.
13. The aerosol provision system of claim 1, wherein the setting is a magnitude of a force generated by the haptic component.
14. The aerosol provision system of claim 1, further comprising a display component, and the control circuitry is further configured to adjust a setting of the display component in response to determining that an inhalation on the aerosol provision system is about to occur.
15. (canceled)
16. The aerosol provision system of claim 14, wherein adjusting the setting of the display component corresponds to disabling the display component.
17. The aerosol provision system of claim 1, further comprising a speaker component, and the control circuitry is further configured to adjust a setting of the speaker component in response to determining that an inhalation on the aerosol provision system is about to occur.
18. (canceled)
19. (canceled)
20. The aerosol provision system of claim 1, further comprising a communications interface, and wherein control circuitry is further configured to adjust a setting of the haptic component based on a signal received via the communications interface from an external device.
21. The aerosol provision system of claim 1, further comprising an input device, and wherein control circuitry is further configured to adjust a setting of the haptic component based on an input received via the input device.
22. An aerosol provision device for an aerosol provision system, the aerosol provision system comprising a haptic component, wherein the aerosol provision device comprises control circuitry configured to: adjust a setting of the haptic component in response to determining that an inhalation on the aerosol provision system is about to occur.
23. A method for operating an aerosol provision system comprising: adjusting a setting of a haptic component of the aerosol provision system in response to determining that an inhalation on the aerosol provision system is about to occur.
24. (canceled)